

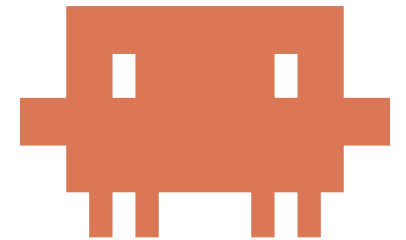
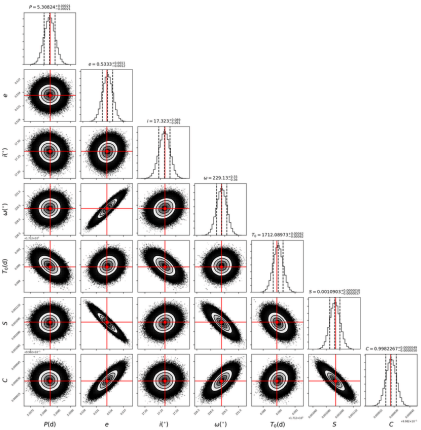
From Supercomputers to AI Agents

Modern workflows for high-performance computing

Tom Kimpson

MACSYS · 2026

Follow along afterwards: github.com/tomkimpson/...



Two skills

Part 1 — Slurm

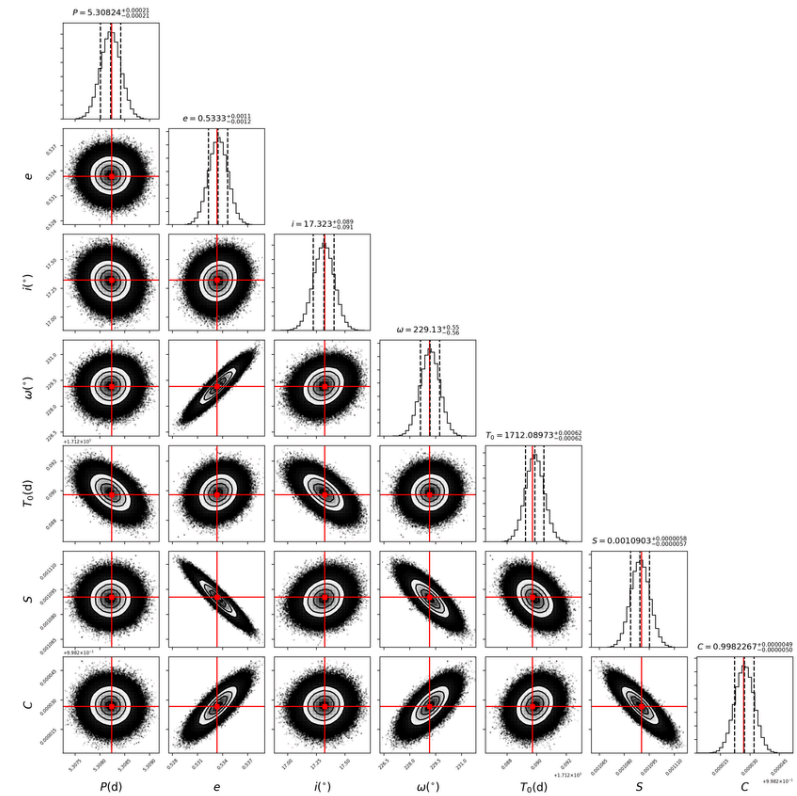
How to run intensive computations on a remote machine

Part 2 — AI agents

How to get an agent to write/debug your code

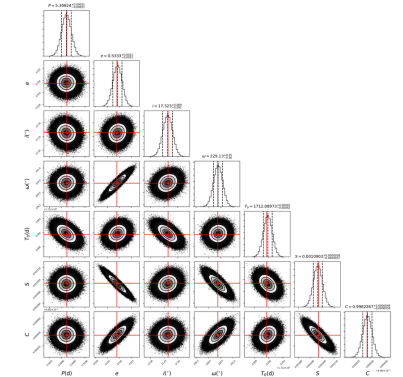
We will be jumping back and forth between these slides and some live terminal demos

Part 1 : HPC + slurm



Why bother?

Lots of science is now compute-intensive and/or scalable



Bottlenecked by your laptop

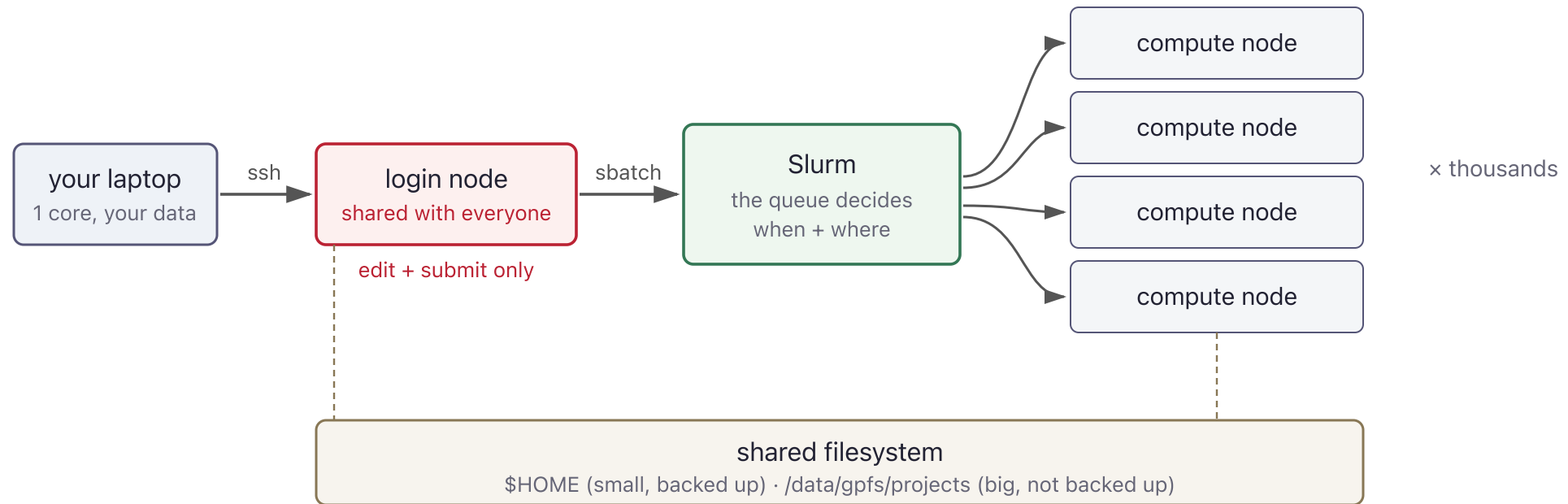
- Your job needs 200 GB of RAM. Your laptop has 16.
- One run takes 6 hours. You need 10,000 runs.
- You closed the lid / it went to sleep / you needed to fly somewhere.
- Your "quick test" has been running since Tuesday and you can't use your machine.

What you get instead

- Hundreds of independent runs at once — parameter sweeps, MCMC chains, bootstrap replicates, seeds
- Nodes with 100s of cores, TB of RAM, GPUs
- Jobs queue and run while you sleep; you get an email when they're done
- Reproducibility for free: the submission script is the record of what you ran

HPC doesn't make your code faster. It lets you run a lot of it at once, on hardware you don't own, without babysitting it.

What a cluster actually is



How to get onto the cluster

Before you can connect

- An account — request via your institution's HPC service desk
- Know your hostname:
`spartan.hpc.unimelb.edu.au` (or whichever cluster)
- On/off campus? Some clusters require VPN from outside
- SSH key pair — generate once, upload the public half:
`ssh-keygen -t ed25519`

Connecting

- **ssh** — the baseline. `ssh user@host`
 - Set up `~/.ssh/config` (or alias) on day one so it's just `ssh spartan`
- **mosh** — ssh that survives suspend, wifi drops, and IP changes
 - Needs mosh installed on the cluster; not all sites allow it
- **tmux** — run *on* the cluster. Sessions persist after you disconnect
 - `tmux new -s work`, detach `Ctrl-b d`, come back `tmux a -t work`
- **VSCode Remote-SSH** — edit files as if local, integrated terminal
 - Caveat: runs a server on the login node, can be resource-hungry. Some sites discourage or block it

How to run a job

You don't run your code directly. Instead you **describe a job** and the scheduler (slurm = Simple Linux Utility for Resource Management) decides when it runs.

Nothing runs on the login node

You tell the scheduler

- How long it will take — `--time`
- How much memory — `--mem`
- How many cores — `--cpus-per-task`
- Which pool of nodes — `--partition`
- Who pays — `--account`

What the scheduler does

- Allocates your job a place in the queue
- Packs small honest jobs in **sooner** than big greedy ones.
- Kills you at `--time`.
- Kills you at `--mem`.

*The queue is not first-come-first-served. It is an optimiser, and your resource request is your bid.
Ask for less, wait less.*

How to describe a job

To run a job you have to write a shell script. A job script is just a shell script. The `#SBATCH` lines are comments to the shell and instructions to the scheduler.

```
#!/bin/bash
#SBATCH --job-name=ou           # what shows up in queue
#SBATCH --account=punimXXXX     # who gets billed
#SBATCH --partition=cascade     # which pool of nodes
#SBATCH --time=00:05:00        # HH:MM:SS – a hard kill, not a hint
#SBATCH --ntasks=1             # one process...
#SBATCH --cpus-per-task=1      # ...on one core
#SBATCH --mem=2G               # for the whole job – also a hard kill
#SBATCH --output=logs/%x-%j.out # %x = job name, %j = job id
#SBATCH --mail-type=END,FAIL    # tell me when it lands, or dies
#SBATCH --mail-user=you@unimelb.edu.au

set -euo pipefail              # fail loudly, not silently

module purge                   # start from a known state
module load foss/2022a Python/3.10.4
source ~/venvs/macsys/bin/activate

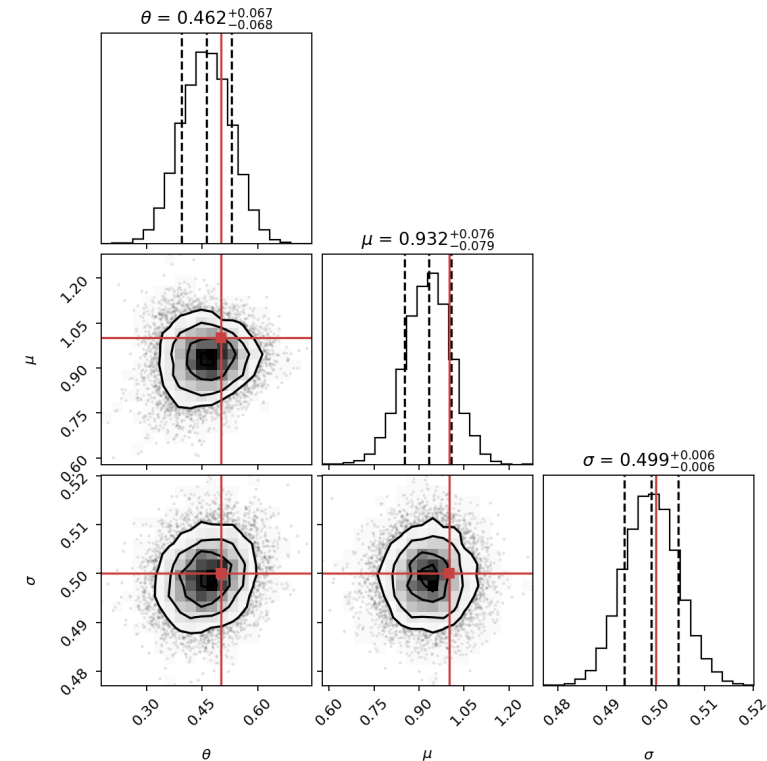
srun python 01_mcmc.py --out posterior.png
```

Basic driving skills

- `sinfo` to see what's available
- `sbatch` submits your job
- `sinteractive` for a shell on a compute node — debug there, never on the login node
- `queue --me` to see your jobs, and why they're waiting
- `scancel $ID` to kill one
- `sacct -j $ID` for what happened to it — `queue` forgets finished jobs
- `seff $ID` for how much you actually used

Example job monitor from Swinburne

e.g. `sbatch 01_run.sh`



One job → one hundred

An **array job** is the same script run N times.

```
#SBATCH --array=0-99%20      # 100 tasks, 20 at once

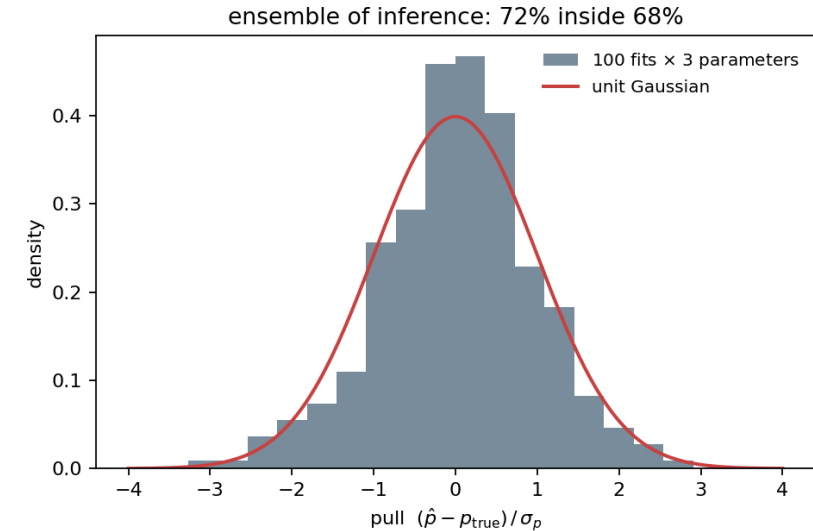
srun python 02_inference.py \
    --task-id ${SLURM_ARRAY_TASK_ID} \
    --out results/fit_${SLURM_ARRAY_TASK_ID}.npz
```

- `--time` and `--mem` are **per task**, not for the array.
- `%20` throttles you. An unthrottled 5,000-task array makes you unpopular!
- Each task writes its own file.

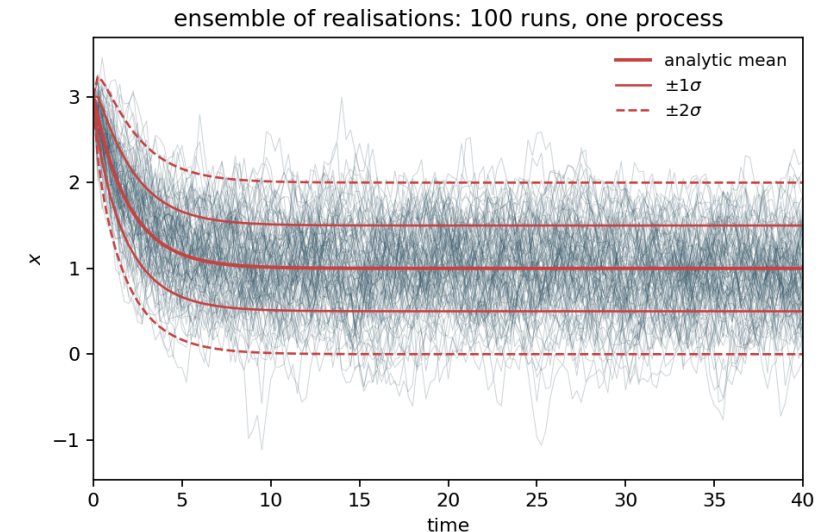
```
sbatch 02_sweep.sh
queue --me
sbatch 02_collate.sh
```

Slurm can hold the collate job until the array finishes — see the appendix.

02_sweep.sh — 100 MCMC fits



03_sweep.sh — 100 forward simulations



Gotchas

	The mistake	The fix
1	Running the real thing on the login node	<code>sinteractive</code> for debugging. That's what it's for.
2	<code>--time</code> too short → job killed at 99%	Time a short version first, then ask for ~1.5×
3	<code>--time</code> / <code>--mem</code> too <i>long</i> → queued for hours	<code>seff</code> after every job. Bring the numbers down.
4	Results written to <code>\$HOME</code>	Small and backed up. Data goes in project storage.
5	Environment differs between login and compute	<code>module purge</code> first, then load explicitly, in the script

Cheat sheet

Getting on and getting oriented

```
ssh spartan.hpc.unimelb.edu.au
module spider python          # search
module load foss/2022a Python/3.10.4
module list                   # what's loaded
sinfo -s                      # partitions, right now
```

Submitting

```
sbatch 01_run.sh
sbatch --parsable 02_sweep.sh      # id only
sbatch --dependency=afterok:$ID 02_collate.sh
sinteractive --time=1:00:00 --mem=8G
```

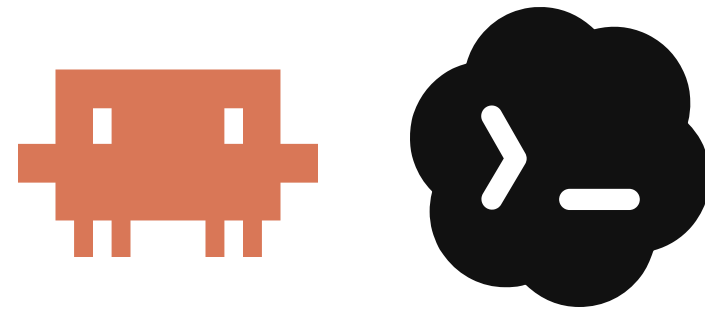
Watching

```
squeue --me
squeue --start -j $ID           # why am I waiting?
scontrol show job $ID
tail -f logs/ou-fit-$ID.out
```

Afterwards

```
seff $ID                        # did I size it right?
sacct -j $ID --format=JobID,State,Elapsed,MaxRSS,ReqMem
scancel $ID                     # one job
scancel ${ID}_7                 # one array task
```

Part 2 : AI agents



How most people use AI already

1. The chat window

ChatGPT / Claude / Gemini in a browser tab.

You paste in the traceback. It pastes back a fix. You paste that into your editor and hope you copied all of it.

2. Tab-complete

Copilot's grey ghost text in VSCode.

It guesses the next line or two from what is above the cursor. You hit **Tab**.

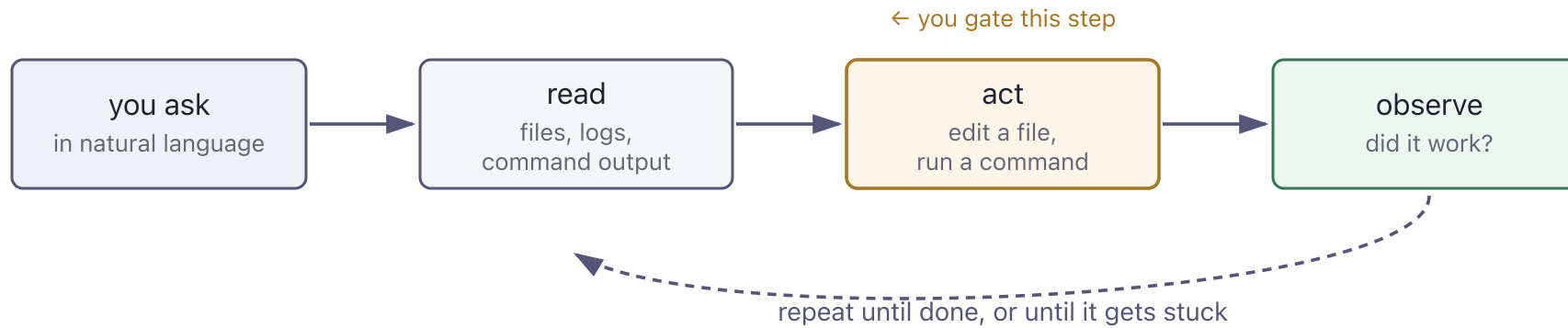
3. Inline edit

Highlight a function, **Cmd-K**, *"vectorise this"*.

The same conversation as (1), but it can at least see the file you are in.

AI agent

model + tools + loop



The landscape



Claude Code

Anthropic. Terminal-first, also in the IDE.



Codex

OpenAI. Terminal and cloud.



Antigravity CLI

Google. Terminal only (?).

+ IDE-native agents (Cursor, Windsurf, Copilot agent mode) and open-source harnesses (Aider, OpenHands) — though an open *harness* almost always still calls a closed *model*. On how far the open-weight models actually trail: [How far behind are open models?](#)

The differences are real but small, and they move every few weeks.

Claude Code: basic driving skills

Starting out

<code>/init</code>	Write a starter <code>CLAUDE.md</code> for the repo
<code>/memory</code>	Edit <code>CLAUDE.md</code> afterwards
<code>/permissions</code>	What it may run without asking
<code>/doctor</code>	Setup checkup; also trims a bloated <code>CLAUDE.md</code>

Steering a task

<code>/plan</code>	Plan first, act second. Use before anything large
<code>/diff</code>	What has it actually changed?
<code>/rewind</code>	Roll code and conversation back to a checkpoint

Context is the state

<code>/context</code>	What is filling the window
<code>/compact</code>	Summarise to free space, same conversation
<code>/clear</code>	New conversation, empty context

Before you ship

<code>/simplify</code>	Cleanup pass on what it just wrote: duplication, nesting, naming. Claims to preserve behaviour — so only run it with tests in place
<code>/code-review</code>	Reviews the diff for correctness bugs. <code>--fix</code> applies the findings
<code>/batch ;</code>	Decomposes a codebase-wide change into 5–30 units, one background agent per unit, each in its own git worktree, each opening a PR

Worth knowing about

`/model` `/effort` · `/usage` (cost) · `/resume` · `/mcp`
· `/btw` (side question, off the record)

demo

1. Lets fix this bug...
2. Lets soup-up our MCMC to use JAX, Hamiltonian Monte Carlo with autodiff, and create a visualisation

Opinions / tips (1/2)

How to drive it

- **Research is not vibe-coding an app.** The litmus test: *would you recognise a wrong answer?*
- **Good software engineering matters more, not less.** Branches, PRs, small commits, unit tests, etc.
- **Keep asks small; /clear often.** Hallucinations, unwanted changes, weakened tests and mostly context problems.
- **Parallel agents/tabs work, but easy to lose track.** Start with one.
- **Bio work often banned from using top models !!!**

Stuff to definitely delegate

- Plotting and figure iteration
- File I/O, format wrangling, data munging
- Test scaffolds around code you already trust
- Mechanical refactors, porting between languages
- Reading an unfamiliar codebase — ask before you edit

Opinions / tips (2/2)

Say that you used it

- **Don't hide it.** Say what you used. Everyone else is using it too!
- **Record which models, and for what:** writing, code, algebra, the maths itself.
- **Say what *you* checked,** and take responsibility for the content.
- **Make every number regenerable.** If each figure and table comes out of a script in the repo, the disclosure writes itself.

The **Leiden Declaration on AI and Mathematics** (June 2026, endorsed by the IMU) recommends a disclosure section leidendeclaration.ai

Declaration on AI usage

Following the disclosure principles of the Leiden Declaration on Artificial Intelligence and Mathematics (Alper et al., 2026), we record the AI tools this work relied on. Large language models (primarily Claude Opus 5, with additional inputs from Claude Fable 5 and GPT-5.6-Sol) were used for writing, refactoring and debugging the benchmark and analysis code. The mathematics was derived without LLM assistance; LLMs were used to check the consistency of the algebra. Every reported number is regenerated from the committed records by scripts in the repository, and the authors take full responsibility for the content of this paper.

*With great power comes great responsibility. **You** are ultimately responsible for the work.*

What it costs

US\$ / month	Claude	Codex (OpenAI)	Antigravity (Google)
Free	0 — <i>no</i> Claude Code	0 — limited	0 — limited
Entry	"Pro" \$20	"Plus" \$20	"AI Pro" \$20
Heavy — 5x / 20x	"Max" \$100 / \$200	"Pro" \$100 / \$200	"AI Ultra" \$100 / \$200

Subscription — flat rate, metered by a rolling **5-hour** window *and* a **weekly** cap.

API key — per token, no ceiling. Opus 5 is ~\$5 / \$25 per million tokens in / out.

For a group:

- **Claude Team**, 5x Max, \$25 / user.
- **Claude Team for Scientists**, 5x Max, \$15/user. See <https://www.anthropic.com/news/expanding-support-for-scientists>
- UoM – see Kate for a licence
- Others – as PI to look at Claude Team for Scientists

Consider budgeting for AI usage on any grant applications.

Take-homes

Slurm

- Don't run intensive compute on your laptop
- Nothing runs on the login node.
- HPC doesn't necessarily make your code faster.
- Your resource request is a bid. Ask for less, wait less.
- Try to judge memory/time reqs. accurately
- The submission script is the record of what you ran.

Agents

- **model + tools + loop** — not a chat window. It reads, runs, and checks its own work.
- **Delegate the mechanical stuff** — plots, I/O, refactors, reading unfamiliar code.
- **The litmus test: would you recognise a wrong answer?**
If not, don't delegate it.
- **Context is the state.** Small asks, `/clear` often.
- **Say that you used it.** You own the result either way.

Thanks

Everything in this talk, ready to clone and run:

```
github.com/tomkimpson/...
```

```
01_mcmc.py · 02_inference.py · 03_ensemble.py · ou.py · CLAUDE.md
```

Spartan docs: dashboard.hpc.unimelb.edu.au · Slurm docs: slurm.schedmd.com · Claude Code: docs.claude.com/claude-code

Appendix

The science in the demo appendix

Ornstein–Uhlenbeck: the canonical noisy relaxation. A quantity is pulled back towards a mean at rate θ and kicked by white noise of amplitude σ .

$$dx = \theta (\mu - x) dt + \sigma dW$$

Started from $x(0) = x_0$, the state is exactly Gaussian at every later time:

$$\langle x(t) \rangle = \mu + (x_0 - \mu) e^{-\theta t}, \quad \text{Var } x(t) = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta t})$$

- The **envelope** in 03 is that formula. Nothing is fitted to draw it.
- The transition density being exact means the *likelihood* is exact too — simulation and inference use the same three lines of algebra, and there is no discretisation error to argue about.
- Nothing exotic: this is Langevin dynamics, and AR(1) in discrete time.

Having an analytic answer is deliberate — **the demo can be checked**, which is exactly the property you want when an agent has been editing your code.

Chaining jobs with `--dependency` appendix

Submit the whole pipeline at once and walk away. Slurm holds each stage until the one before it is done — useful when the array will sit in the queue for six hours.

```
ARRAY=$(sbatch --parsable 02_sweep.sh)          # --parsable prints just "12345679"
sbatch --dependency=afterok:$ARRAY 02_collate.sh
```

Plain `sbatch` prints `Submitted batch job 12345679`; `--parsable` prints the bare id so you can capture it. The collate job queues *immediately*, sits at `PD ... (Dependency)`, and starts by itself when the last task exits cleanly.

flavour	starts when the array...
<code>afterok:\$ID</code>	finished, and every task succeeded
<code>afterany:\$ID</code>	finished, pass or fail
<code>afternotok:\$ID</code>	failed — for cleanup and alerts
<code>aftercorr:\$ID</code>	task <i>i</i> waits for task <i>i</i> , one-to-one
<code>singleton</code>	(one job of this name at a time)

The trap. With `afterok`, one failed task means the dependent job *never runs*. It does not error — it sits at `Reason=(DependencyNeverSatisfied)` until you `scancel` it. Some sites clear it via `kill_invalid_depend`; many do not.

So: `afterok` when a partial result is worthless, `afterany` when your collator can cope with missing files.

`srun` inside `sbatch`, and other things people ask appendix

Why `srun python ...` and not just `python ...`? `sbatch` allocates resources; `srun` launches a *step* inside that allocation. For a single-core job it barely matters, but it gets you per-step accounting in `sacct`, and it is what you need the moment you go multi-node. Harmless habit, useful later.

`--ntasks` vs `--cpus-per-task`? `--ntasks` = how many separate processes (MPI ranks). `--cpus-per-task` = how many cores each one gets (threads, OpenMP). One process on 8 threads is `--ntasks=1 --cpus-per-task=8`. Eight MPI ranks is `--ntasks=8 --cpus-per-task=1`. Most of the time you want neither — you want an array of 100 single-core jobs.

What about GPUs? `--partition=gpu-a100 --gres=gpu:1`. Everything else is the same. The hard part is never Slurm; it is whether your code uses the GPU.

What if a few array tasks fail? `sacct -j $ARRAYID --state=FAILED` lists them, and `sbatch --array=7,23,88 02_sweep.sh` reruns just those. This is why one file per task matters.

Running an agent *on* the cluster appendix

You can install and run Claude Code on a login node, which makes the debugging loop tighter — it reads the logs and resubmits without anything round-tripping to your laptop.

Two caveats worth knowing before you do:

- **It is a shared login node.** Agents run commands. Everything on the etiquette slide applies double, and `CLAUDE.md` should say so explicitly.
- **Check your site's policy on outbound API calls** and on what may leave the cluster. If your data is sensitive, the agent reading your files is a data-governance question, not just a technical one.

The alternative, and what I usually do: run the agent on the laptop in the repo, and let it `ssh` out for the cluster-side commands. Slower loop, clearer boundary.

```
config Host spartan HostName spartan.hpc.unimelb.edu.au User tkimpson ServerAliveInterval 60
```